MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, DTL,

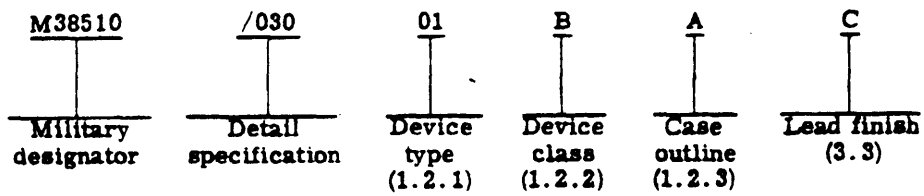
NAND GATES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, DTL, positive logic NAND gating microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.



1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual, 4-input expandable NAND gate
02	Extendable hex inverter
03	Hex inverter
04	Quadruple, 2-input positive NAND gate
05	Triple, 3-input positive NAND gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, MIL-M-38510, appendix C
A	F-1 (14-lead, 1/4" x 1/4", flat-pack)
B	F-3 (14-lead, 1/8" x 1/4", flat-pack)
C	D-1 (14-lead, 1/4" x 3/4", dual-in-line pack)
D	F-2 (14-lead, 1/4" x 3/8", flat-pack)

1.3 Absolute maximum ratings:

Supply voltage range	-0.5 Vdc to 8.0 Vdc
Input voltage range	-1.5 Vdc to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Maximum power dissipation, P_D	23 mWdc per gate
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction to case	θ_{JC} = 0.09°C/mW for flat-pack; 0.08°C/mW for dual-in-line pack
Junction temperature	T_J = 175°C

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1.4 Recommended operating conditions:

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	1.9 Vdc @ 25°C
Maximum low level input voltage	1.1 Vdc @ 25°C
Normalized fanout (each output)	8 maximum
Ambient operating temperature range	-55°C to 125°C

2. APPLICABLE DOCUMENT

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3 herein.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/	Device types	Limits		Units
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -0.12 mA	All	2.85		Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 12 mA	All		0.45	Volts
Low-level input current	I_{IL}	V_{CC} = 5.5 V	All	-0.6	-1.50	mA
High-level input current	I_{IH}	V_{CC} = 5.5 V	01, 03, 04, 05	0	5.0	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{OUT} = 0 V, V_{IN} = GND 2/	All	-0.59	-1.34	mA
High-level supply current per gate	I_{CCH1}	V_{CC} = 5.0 V, V_{IN} = GND	All		1.47	mA
High-level supply current per gate	I_{CCH2}	V_{CC} = 8.0 V, V_{IN} = GND	All		2.75	mA
Low-level supply current per gate	I_{CCL}	V_{CC} = 4.5 V, V_{IN} = OPEN	All		3.25	mA
Collector cutoff current	I_{CEX}	V_{CC} = 4.5 V	All		100	µA
Propagation delay time low-to-high level output	t_{PLH}	V_{CC} = 5.0 V, C_L = 30 pF R_L = 3.9 kΩ	All	25	112	ns
Propagation delay time high-to-low level output	t_{PHL}	V_{CC} = 5.0 V, C_L = 50 pF R_L = 390Ω	01, 03, 04, 05	10	40	ns
			02	10	45	ns

1/ Complete terminal conditions shall be as specified in table III.
2/ Not more than one output should be shorted at a time.

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3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

(a) Country of origin.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)		
	Class A devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (Method 5004)	1	1	None
Final electrical test parameters (Method 5004)	1*, 2, 3, 9	1*, 2, 3, 9	1
Group A test requirements (Method 5005)	1, 2, 3, 9, 10, 11	1, 2, 3, 9	1, 2, 3, 9
Group C end point electrical parameters (Method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for group C periodic inspections	None	10, 11	None

* PDA applies to subgroup 1 (see 4.3(h)).

4. PRODUCT ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and Method 5005 of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, and C inspections (see 4.4.1, 4.4.2, and 4.4.3). After qualification of one or more electrically and structurally similar types with a single lead finish, other lead finishes of the same case outline may be qualified by submitting a single type in the qualified case outline to the group B, subgroup 3 test and the group C, subgroups 1, 3, and 4 tests.

4.3 Screening. Screening shall be in accordance with Method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

(a) Test samples for the group B bond strength test specified in Method 5005 of MIL-STD-883 may, at the manufacturer's option, be randomly selected immediately following the internal visual (precap) inspection and prior to sealing (see 4.4.2(b)).
(b) Temperature cycling (Method 1010 of MIL-STD-883).
    (1) Omit seal test as post-test measurement.

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(c) Thermal shock (Method 1011 of MIL-STD-883), when substituted for temperature cycling.
   (1) Omit seal test as post-test measurement.
(d) Burn-in test (Method 1015 of MIL-STD-883).
   (1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
   (2) T_A = 125°C minimum.
(e) Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of Method 5004 of MIL-STD-883 may be omitted.
(f) Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
(g) External visual inspection shall not include measurement of case and lead dimensions.
(h) Percent defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with Method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of Method 5005 of MIL-STD-883 and as follows:

(a) Tests shall be as specified in table II.
(b) Subgroups 4, 5, 6, 7, and 8 of table I of Method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of Method 5005 of MIL-STD-883 and as follows:

(a) Bond strength test may be conducted on samples collected prior to sealing (see 4.3(a)).

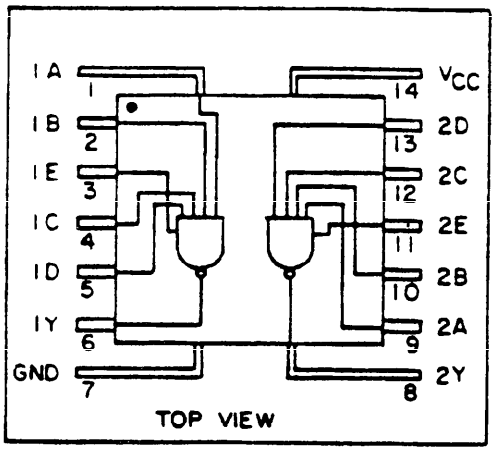
4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Subgroups 7 and 8 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions and limits specified for subgroups 10 and 11 of group A.
(c) Lead bend in only one direction is required for initial conditioning prior to moisture resistance and salt atmosphere tests.
(d) High temperature storage test (Method 1008 of MIL-STD-883) conditions:
   (1) Temperature: 150° ±10°C.
   (2) Duration: 1,000 hours, except as otherwise permitted by Appendix B of MIL-M-38510.
(e) Operating life test (Method 1005 of MIL-STD-883) conditions:
   (1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
   (2) T_A = 125°C, minimum.
   (3) Test duration: 1,000 hours, except as permitted by Appendix B or MIL-M-38510.
(f) Omit steady state reverse bias test.

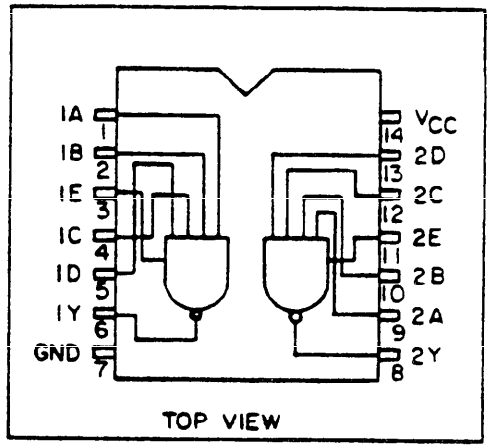
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Device type 01

Cases A, B, and D

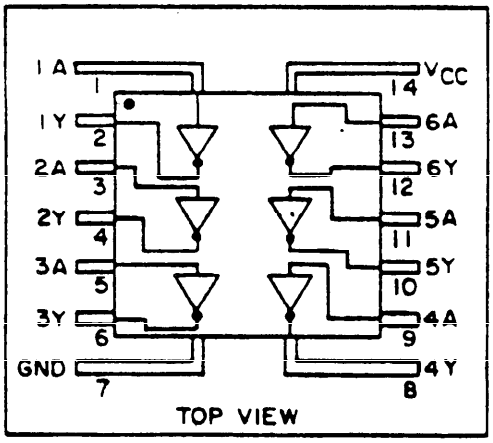


Case C

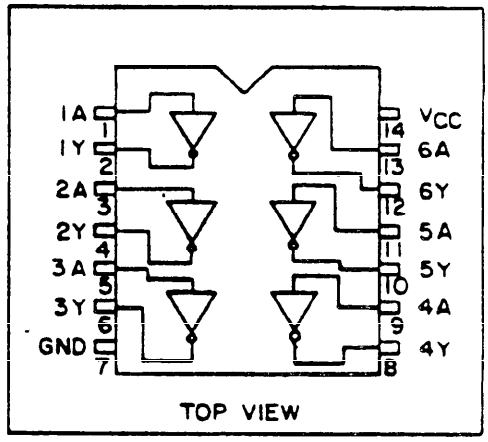


Device types 02 and 03

Cases A, B, and D

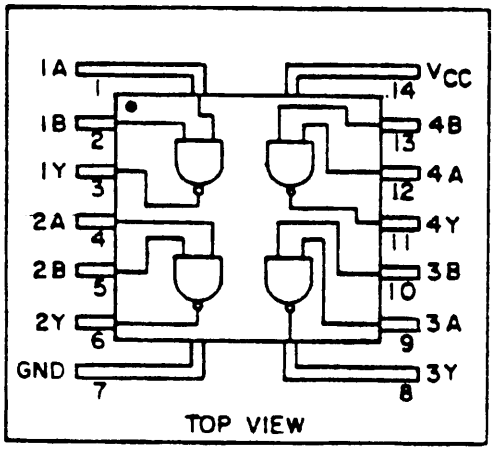


Case C



Device type 04

Cases A, B, and D



Case C

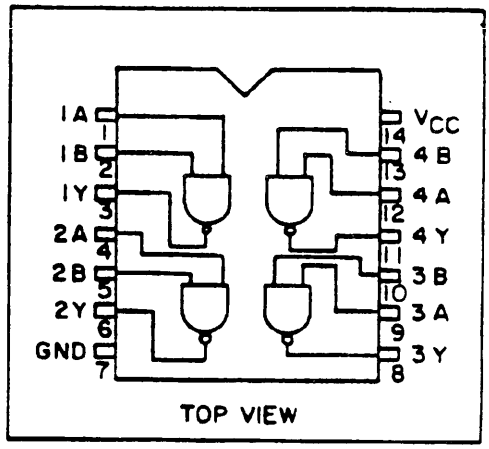
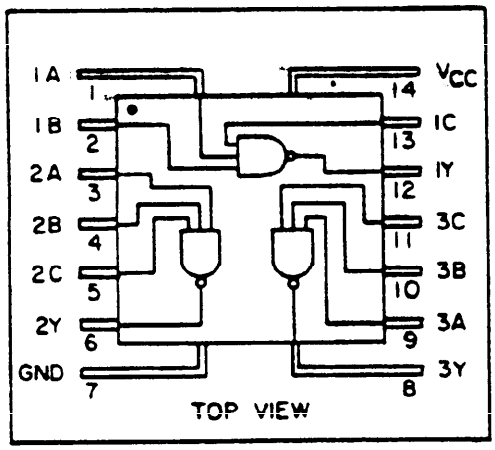


FIGURE 1. Logic diagram and terminal connections.

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Device type 05

Cases A, B, and D



Case C

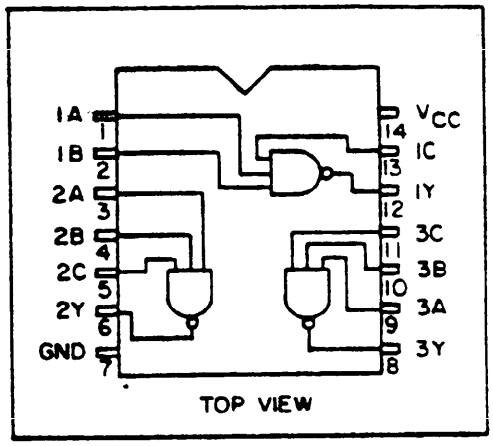


FIGURE 1. Logic diagram and terminal connections (Continued)

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DEVICE TYPE 04

TRUTH TABLE EACH GATE		
INPUT		OUTPUT
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

Positive Logic: $Y = \overline{AB}$

DEVICE TYPE 05

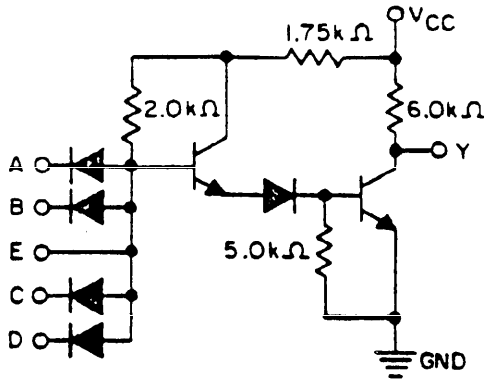
TRUTH TABLE EACH GATE			
INPUT			OUTPUT
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L

Positiv logic: $Y = \overline{A\,B\,C}$

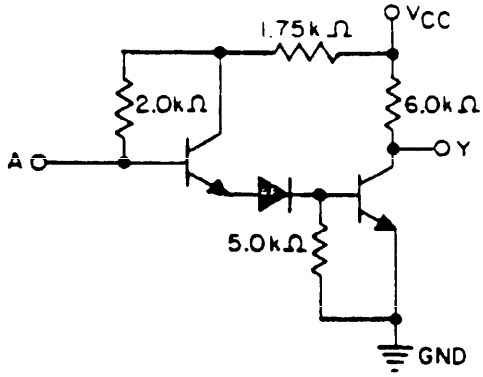
FIGURE 2. Truth tables (continued)

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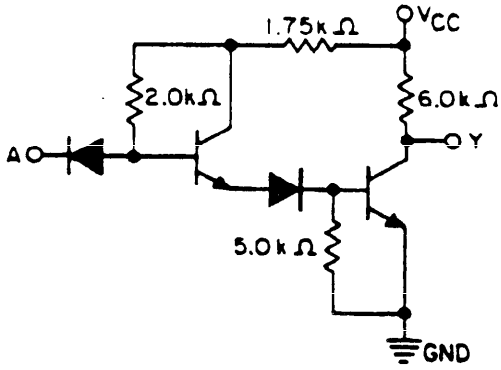
Device type 01
(each gate)



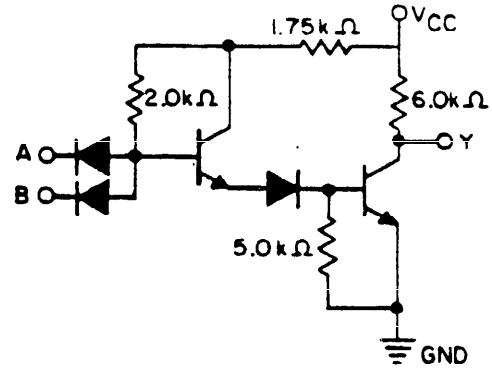
Device type 02
(each inverter)



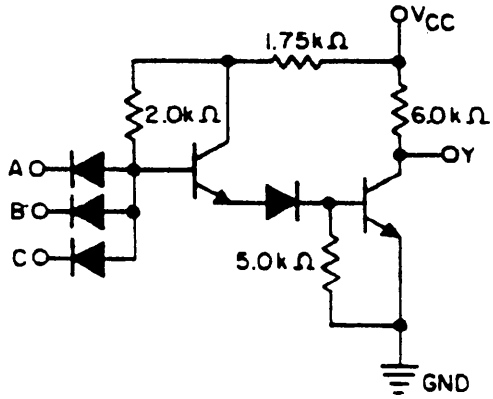
Device type 03
(each inverter)



Device type 04
(each gate)



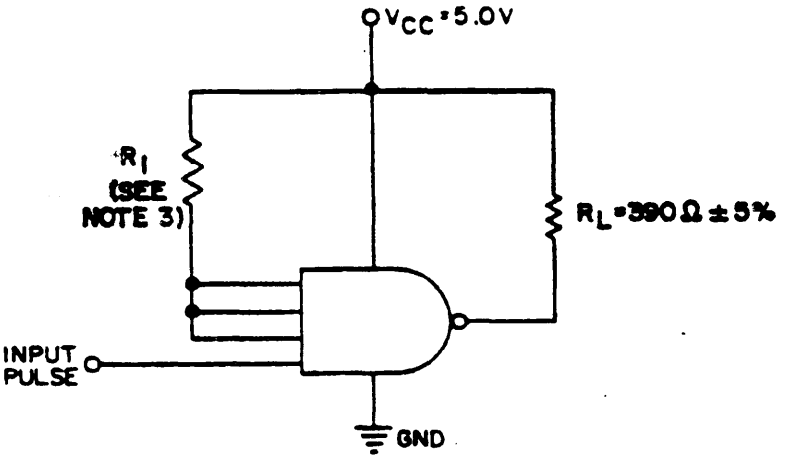
Device type 05
(each gate)



NOTE: All resistance value shown are nominal

FIGURE 3. Schematic circuits.

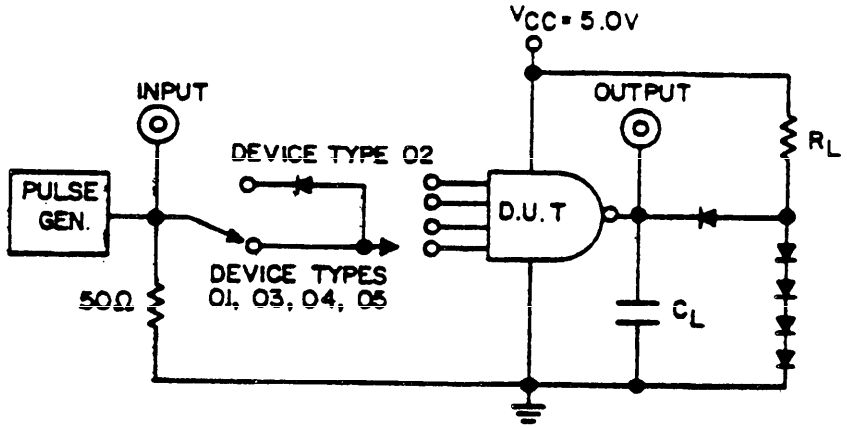
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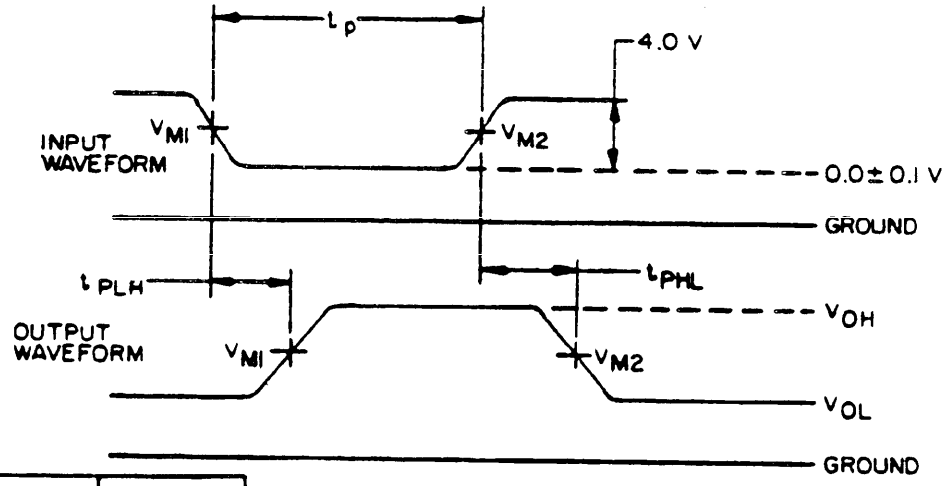
NOTES:
1. The input pulse characteristics are: Vgen = 4.0 V; PRR = 100 kHz; duty cycle = 50%; t_{TLH} = 100 ns; t_{THL} = 100 ns.
2. All input adjustments are ±5%.
3. All undriven inputs are connected to V_{CC} thru appropriate current limiting resistors.
4. For parallel excitation: Clock pulse shall be 1 MHz and undriven inputs shall be connected to V_{CC} thru appropriate current limiting resistors.
5. External input diodes shall be used with device type 02.

FIGURE 4. Burn-in and life test circuit.

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	t_{PLH}	t_{PHL}
R_L	3.9k Ω	390Ω ± 5%
C_L	30 pF min	50 pF min



TEMP (°C)	V_{M1} (volts)	V_{M2} (volts)
25	1.5	1.5
125	1.2	1.2
-55	1.7	1.7

NOTES:
1. The generator has the following characteristics: Vgen = 4.0 V; t_{TLH} = 10 ns ±90%, t_{THL} = 10 ns ±90%; PRR = 500 kHz, t_p = 1 μs.
2. All diodes are 1N4150 or equivalent.
3. CL includes all jig and probe capacitance.

FIGURE 5. Switching time test circuit.

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TABLE III. Group A inspection for device type 01. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
				1A	1B	1E	1C	1D	1Y	GND	2Y	2A	2B	2E	2C	2D	V_{CC}				
1 T_A=25°C	V_{OL}	3007	1	1.8 v	1.8 v		1.8 v	1.8 v	12 mA	GND							4.5 v	1Y		0.4	Vdc
	V_{OL}	3007	2								12 mA	1.8 v	1.8 v		1.8 v	1.8 v		2Y		0.4	
	V_{OH}	3006	3	1.1 v					-0.12mA									1Y	3.4		
			4		1.1 v													1Y			
			5				1.1 v											1Y			
			6					1.1 v										1Y			
			7								-0.12mA	1.1 v						2Y			
			8										1.1 v					2Y			
			9												1.1 v			2Y			
			10													1.1 v		2Y			
	I_{OS}	3011	11	GND					GND								5.5 v	1Y	-.67	-1.34	mA
	I_{OS}	3011	12								GND					GND		2Y	-.67	-1.34	mA
	I_{IH}	3010	13	4.0 v	GND		GND	GND										1A		2.0	µA
			14	GND	4.0 v		GND	GND										1B			
			15	GND	GND		4.0 v	GND										1C			
			16	GND	GND		GND	4.0 v										1D			
			17									4.0 v	GND		GND	GND		2A			
			18									GND	4.0 v		GND	GND		2B			
			19									GND	GND		4.0 v	GND		2C			
			20												GND	4.0 v		2D			
	I_{IL}	3009	21	0.4 v	4.0 v		4.0 v	4.0 v										1A	-0.6	-1.60	mA
			22	4.0 v	0.4 v		4.0 v											1B			
			23		4.0 v		0.4 v											1C			
			24		4.0 v		4.0 v	0.4 v										1D			
			25									0.4 v	4.0 v		4.0 v	4.0 v		2A			
			26									4.0 v	0.4 v		4.0 v			2B			
			27										4.0 v		4.0 v			2C			
			28													0.4 v		2D			
	I_{CEX}		29	1.1 v					4.5 v								4.5 v	1Y		50	µA
	I_{CEX}		30								4.5 v				1.1 v		4.5 v	2Y		50	µA
	I_{CCL}	3005	31														5.0 v	VCC		0.5	mA
	I_{CCH1}		32	GND												GND	5.0 v	VCC		2.64	mA
	I_{CCH2}		33	GND												GND	5.0 v	VCC		5.5	mA
	$V_{OH(E)}$	3006	34			1.8 v			-.12 mA								4.5 v	1Y	3.4		Vdc
	$V_{OH(R)}$	3006	35								-.12 mA			1.8 v				2Y	3.4		Vdc
2 T_A=125°C	V_{OL}	3007	36	1.7 v	1.7 v		1.7 v	1.7 v	12 mA									1Y		0.45	Vdc
	V_{OL}	3007	37								12 mA	1.7 v	1.7 v		1.7 v	1.7 v		2Y		0.45	Vdc
	V_{OH}	3006	38	0.8 v					-0.12mA									1Y	2.85		
			39		0.8 v													1Y			
			40				0.8 v											1Y			
			41					0.8 v										1Y			
			42								-.12 mA	0.8 v						2Y			
			43										0.8 v					2Y			
			44												0.8 v			2Y			
			45													0.8 v		2Y			
	I_{OS}	3011	46	GND					GND								5.5 v	1Y	-.59	-1.3	mA
	I_{OS}	3011	47								GND					GND	5.5 v	2Y	-.59	-1.3	mA

See footnote at end of table.

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TABLE III. Group A inspection for device type 01 - Continued

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1B	3 1E	4 1C	5 1D	6 1Y	7 GND	8 2Y	9 2A	10 2B	11 2E	12 2C	13 2D	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
2 T_A=125°C	I_{IH}	3010	48	4.0 v	GND		GND	GND		GND							5.5 v	1A		5.0	µA
			49	GND	4.0 v		GND											1B			
			50		GND		4.0 v											1C			
			51		GND		GND	4.0 v										1D			
			52									4.0 v	GND		GND	GND		2A			
			53									GND	4.0 v		GND			2B			
			54										GND		4.0 v			2C			
			55										GND		GND	4.0 v		2D			
	I_{IL}	3009	56	0.4 v	4.0 v		4.0 v	4.0 v										1A	-.6	-1.50	mA
			57	4.0 v	0.4 v		4.0 v											1B			
			58		4.0 v		0.4 v											1C			
			59		4.0 v		4.0 v	0.4 v										1D			
			60									0.4 v	4.0 v		4.0 v	4.0 v		2A			
			61									4.0 v	0.4 v		4.0 v			2B			
			62										4.0 v		0.4 v			2C			
			63										4.0 v		4.0 v	0.4 v		2D			
	I_{CEX}		64	0.8 v					4.5 v								4.5 v	1Y		100	µA
	I_{CEX}		65								4.5 v				0.8 v		4.5 v	2Y		100	µA
	I_{CCL}	3005	66														5.0 v	V_{CC}		0.5	mA
	I_{CCH1}		67	GND												GND	5.0 v	V_{CC}		2.94	mA
	I_{CCH2}		68	GND												GND	5.0 v	V_{CC}		5.5	mA
	$V_{OH(E)}$	3006	69			1.3 v			-.12 mA								4.5 v	1Y	3.3		Vdc
	$V_{OH(E)}$	3006	70								-.12 mA			1.3 v				2Y	3.3		
3 T_A= -55°C	V_{OL}	3007	71	2.1 v	2.1 v		2.1 v	2.1 v	12 mA									1Y		0.4	
	V_{OL}	3007	72								12 mA	2.1 v	2.1 v		2.1 v	2.1 v		2Y		0.4	
	V_{OH}	3006	73	1.4 v					-0.12mA									1Y	3.3		
			74		1.4 v													1Y			
			75				1.4 v											1Y			
			76					1.4 v										1Y			
			77								-.12 mA	1.4 v						2Y			
			78										1.4 v					2Y			
			79												1.4 v			2Y			
			80													1.4 v		2Y			
	I_{OS}	3011	81	GND					GND								5.5 v	1Y	-.67	-1.34	mA
	I_{OS}	3011	82								GND					GND		2Y	-.67	-1.34	mA
	I_{IH}	3010	83	4.0 v	GND		GND	GND										1A		2.0	µA
			84	GND	4.0 v		GND											1B			
			85		GND		4.0 v											1C			
			86		GND		GND	4.0 v										1D			
			87									4.0 v	GND		GND	GND		2A			
			88									GND	4.0 v		GND			2B			
			89										GND		4.0 v			2C			
			90										GND		GND	4.0 v		2D			
	I_{IL}	3009	91	0.4 v	4.0 v		4.0 v	4.0 v										1A	-.6	-1.50	mA
			92	4.0 v	0.4 v		4.0 v											1B			
			93		4.0 v		0.4 v											1C			
			94		4.0 v		4.0 v	0.4 v										1D			
			95									0.4 v	4.0 v		4.0 v	4.0 v		2A			
			96									4.0 v	0.4 v		4.0 v			2B			
			97										4.0 v		0.4 v			2C			
			98										4.0 v		4.0 v	0.4 v		2D			

See footnotes at end of t

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TABLE III. Group A inspection for device type 01 - Continued

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1B	3 1E	4 1C	5 1D	6 1Y	7 GND	8 2Y	9 2A	10 2B	11 2E	12 2C	13 2D	14 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Unit
3 T_A= -55°C	I_{CEX}		99	1.4 v					4.5 v	GND							4.5 v	1Y		50	µA
	I_{CEX}		100								4.5 v					1.4 v	4.5 v	2Y		50	µA
	I_{CCL}	3005	101														5.0 v	V_{CC}		6.5	mA
	I_{CCH1}		102	GND												GND	5.0 v	V_{CC}		2.94	mA
	I_{CCH2}		103	GND												GND	5.0 v	V_{CC}		5.5	mA
	$V_{OH(E)}$	3006	104			1.1 v			-.12 mA					3.1 v			4.5 v	1Y	3.3		Vdc
	$V_{OH(E)}$	3006	105								-.12 mA						4.5 v	2Y	3.3		Vdc
9 T_A=25°C	t_{PHL}	3003	106	IN					OUT								5.0 v	1A to 1Y	10	30	ns
	t_{PHL}	(Fig. 5)	107								OUT	IN						2A to 2Y	10	30	
	t_{PLH}		108	IN					OUT									1A to 1Y	25	80	
	t_{PLH}		109								OUT	IN						2A to 2Y	25	80	
10 T_A=125°C	t_{PHL}		110	IN					OUT									1A to 1Y	10	35	
	t_{PHL}		111								OUT	IN						2A to 2Y	10	35	
	t_{PLH}		112	IN					OUT									1A to 1Y	25	112	
	t_{PLH}		113								OUT	IN						2A to 2Y	25	112	
11 T_A= -55°C	t_{PHL}		114	IN					OUT									1A to 1Y	10	40	
	t_{PHL}		115								OUT	IN						2A to 2Y	10	40	
	t_{PLH}		116	IN					OUT									1A to 1Y	25	80	
	t_{PLH}		117								OUT	IN						2A to 2Y	25	80	

1/ Pins not designated may be HIGH level logic, LOW level logic or open.

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TABLE III. Group A inspection for device type 02. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1Y	3 2A	4 2Y	5 3A	6 3Y	7 GND	8 4Y	9 4A	10 5Y	11 5A	12 6Y	13 6A	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A=25°C	V_{OL}	3007	1	2.6 v	12 mA	2.6 v	12 mA	2.6 v	12 mA	GND	12 mA	2.6 v	12 mA	2.6 v	12 mA	2.6 v	4.5 v	1Y		0.4	Vdc
			2															2Y			
			3															3Y			
			4															4Y			
			5															5Y			
			6															6Y			
	V_{OH}	3006	7	1.7 v	-0.12mA	1.7 v	-0.12mA	1.7 v	-0.12mA		-0.12mA	1.7 v	-0.12 mA	1.7 v	-0.12 mA	1.7 v		1Y	3.4		
			8															2Y			
			9															3Y			
			10															4Y			
			11															5Y			
			12															6Y			
	I_{OS}	3011	13	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	5.5 v	1Y	-.67	-1.34	mA
			14															2Y			
			15															3Y			
			16															4Y			
			17															5Y			
			18															6Y			
	I_{IL}	3009	19	1.0 v		1.0 v		1.0 v				1.0 v		1.0 v		1.0 v		1A	-.8	-1.50	mA
			20															2A			
			21															3A			
			22															4A			
			23															5A			
			24															6A			
	I_{CEX}		25	4.5 v	4.5 v	4.5 v	4.5 v	4.5 v	4.5 v		4.5 v	4.5 v		4.5 v		4.5 v	4.5 v	1Y		50	µA
			26															2Y			
			27															3Y			
			28															4Y			
			29															5Y			
			30															6Y			
	I_{CCL}	3005	31														5.0 v	V_{CC}		19.5	mA
	I_{CCH1}		32	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		8.82	mA
	I_{CCH2}		33	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		16.5	mA
2 T_A=125°C	V_{OL}	3007	34	2.21 v	12 mA	2.21 v	12 mA	2.21 v	12 mA		12 mA	2.21 v	12 mA	2.21 v	12 mA	2.21 v	4.5 v	1Y		0.45	Vdc
			35															2Y			
			36															3Y			
			37															4Y			
			38															5Y			
			39															6Y			
	V_{OH}	3006	40	1.21 v	-0.12mA	1.21 v	-0.12mA	1.21 v	-0.12mA		-0.12mA	1.21 v	-0.12 mA	1.21 v	-0.12mA	1.21 v		1Y	2.85		
			41															2Y			
			42															3Y			
			43															4Y			
			44															5Y			
			45															6Y			

See footnote at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1Y	3 2A	4 2Y	5 3A	6 3Y	7 GND	8 4Y	9 4A	10 5Y	11 5A	12 6Y	13 6A	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
2 T_A=125°C	I_{OS}	3011	46	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 v	1Y	-.59	-1.3	mA
			47															2Y			
			48															3Y			
			49															4Y			
			50															5Y			
			51															6Y			
	I_{IL}	3009	52	0.81 v		0.81 v		0.81 v				0.81 v		0.81 v		0.81 v		1A	-.6	-1.50	mA
			53															2A			
			54															3A			
			55															4A			
			56															5A			
			57															6A			
	I_{CEX}		58	1.21 v	4.5 v	1.21 v	4.5 v	1.21 v	4.5 v		4.5 v	1.21 v	4.5 v	1.21 v	4.5 v	1.21 v	4.5 v	1Y		100	µA
			59															2Y			
			60															3Y			
			61															4Y			
			62															5Y			
			63															6Y			
	I_{CCL}	3005	64														5.0 v	V_{CC}		19.5	mA
	I_{CCH1}		65	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		8.82	mA
	I_{CCH2}		66	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		16.5	mA
3 T_A = -55°C	V_{OL}	3007	67	3.02 v	12 mA	3.02 v	12 mA	3.02 v	12 mA		12 mA	3.02 v	12 mA	3.02 v	12 mA	3.02 v	4.5 v	1Y		0.4	Vdc
			68															2Y			
			69															3Y			
			70															4Y			
			71															5Y			
			72															6Y			
	V_{OH}	3006	73	2.32 v	-0.12mA	2.32 v	-0.12mA	2.32 v	-0.12mA		-0.12mA	2.32 v	-0.12mA	2.32 v	-0.12 mA	2.32 v		1Y	3.3		
			74															2Y			
			75															3Y			
			76															4Y			
			77															5Y			
			78															6Y			
	I_{OS}	3011	79	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	5.5 v	1Y	-.67	-1.34	mA
			80															2Y			
			81															3Y			
			82															4Y			
			83															5Y			
			84															6Y			

See footnote at end of table.

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1Y	3 2A	4 2Y	5 3A	6 3Y	7 GND	8 4Y	9 4A	10 5Y	11 5A	12 6Y	13 6A	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_A= -55°C	I_{IL}	3009	85 86 87 88 89 90	1.16 v		1.16 v		1.16 v		GND		1.16 v		1.16 v		1.16 v	5.5 v	1A 2A 3A 4A 5A 6A	-.6	-1.60	mA
	I_{CEX}		91 92 93 94 95 96	2.32 v	4.5 v	2.32 v	4.5 v	2.32 v	4.5 v		4.5 v	2.32 v	4.5 v	2.32 v	4.5 v	2.32 v	4.5 v	1Y 2Y 3Y 4Y 5Y 6Y		50	µA
	I_{CCL}	3005	97														5.0 v	V_{CC}		19.5	mA
	I_{CCH1}		98	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		6.02	mA
	I_{CCH2}		99	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		16.5	mA
9 T_A=25°C	t_{PHL}	3003 (Fig. 5)	100 101 102 103 104 105	IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	IN	OUT	IN	5.0 v	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y	10	35	ns
	t_{PLH}		106 107 108 109 110 111	IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	IN	OUT	IN		1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y	25	80	
10 T_A=125°C	t_{PHL}		112 113 114 115 116 117	IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	IN	OUT	IN		1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y	10	40	
	t_{PLH}		118 119 120 121 122 123	IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	IN	OUT	IN		1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y	25	112	
11 T_A= -55°C	t_{PHL}	3003	124 125 126 127 128 129	IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	IN	OUT	IN		1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y	10	45	
	t_{PLH}		130 131 132 133 134 135	IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	IN	OUT	IN		1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y	25	80	

1/ Pins not designated may be HIGH level logic, LOW level logic or open.

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TABLE III. Group A inspection for device type 03. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1Y	3 2A	4 2Y	5 3A	6 3Y	7 GND	8 4Y	9 4A	10 5Y	11 5A	12 6Y	13 6A	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{OL}	3007	1	1.9 v	12 mA	1.9 v	12 mA	1.9 v	12 mA	GND	12 mA	1.9 v	12 mA	1.9 v	12 mA	1.9 v	4.5 v	1Y		0.4	Vdc
			2															2Y			
			3															3Y			
			4															4Y			
			5															5Y			
			6															6Y			
	V_{OH}	3006	7	1.1 v	-0.12mA	1.1 v	-0.12mA	1.1 v	-0.12mA		-0.12mA	1.1 v	-0.12mA	1.1 v	-0.12mA	1.1 v		1Y	3.4		
			8															2Y			
			9															3Y			
			10															4Y			
			11															5Y			
			12															6Y			
	I_{OS}	3011	13	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	5.5 v	1Y	-.67	-1.34	mA
			14															2Y			
			15															3Y			
			16															4Y			
			17															5Y			
			18															6Y			
	I_{IH}	3010	19	4.0 v		4.0 v		4.0 v				4.0 v		4.0 v		4.0 v		1A		2	µA
			20															2A			
			21															3A			
			22															4A			
			23															5A			
			24															6A			
	I_{IL}	3009	25	0.4 v		0.4 v		0.4 v				0.4 v		0.4 v		0.4 v		1A	-.6	-1.60	mA
			26															2A			
			27															3A			
			28															4A			
			29															5A			
			30															6A			
	I_{CEX}		31	1.1 v	4.5 v	1.1 v	4.5 v	1.1 v	4.5 v		4.5 v	1.1 v	4.5 v	1.1 v	4.5 v	1.1 v	4.5 v	1Y		50	µA
			32															2Y			
			33															3Y			
			34															4Y			
			35															5Y			
			36															6Y			
	I_{CCL}	3005	37														5.0 v	V_{CC}		19.5	mA
	I_{CCH1}		38	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		8.82	mA
	I_{CCH2}		39	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		16.5	mA
2 T_A = 125°C	V_{OL}	3007	40	1.7 v	12 mA	1.7 v	12 mA	1.7 v	12 mA		12 mA	1.7 v	12 mA	1.7 v	12 mA	1.7 v	4.5 v	1Y		0.45	Vdc
			41															2Y			
			42															3Y			
			43															4Y			
			44															5Y			
			45															6Y			
	V_{OH}	3006	46	0.8 v	-0.12mA	0.8 v	-0.12mA	0.8 v	-0.12mA		-0.12mA	0.8 v	-0.12mA	0.8 v	-0.12mA	0.8 v		1Y	2.85		
			47															2Y			
			48															3Y			
			49															4Y			
			50															5Y			
			51															6Y			

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1Y	3 2A	4 2Y	5 3A	6 3Y	7 GND	8 4Y	9 4A	10 5Y	11 5A	12 6Y	13 6A	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
2 T_A = 125°C	I_{OS}	3011	52 53 54 55 56 57	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	5.5 v	1Y 2Y 3Y 4Y 5Y 6Y	-.59	-1.3	mA
	I_{IH}	3010	58 59 60 61 62 63	4.0 v		4.0 v		4.0 v				4.0 v		4.0 v		4.0 v		1A 2A 3A 4A 5A 6A		5.0	µA
	I_{IL}	3009	64 65 66 67 68 69	0.4 v		0.4 v		0.4 v				0.4 v		0.4 v		0.4 v		1A 2A 3A 4A 5A 6A	-.8	-1.50	mA
	I_{CEX}		70 71 72 73 74 75	0.8 v	4.5 v	0.8 v	4.5 v	0.8 v	4.5 v		4.5 v	0.8 v	4.5 v	0.8 v	4.5 v	0.8 v	4.5 v	1Y 2Y 3Y 4Y 5Y 6Y		100	µA
	I_{CCL}	3005	76														5.0 v	V_{CC}		19.5	mA
	I_{CCH1}		77	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		15.82	mA
	I_{CCH2}		78	GND		GND		GND				GND		GND		GND	5.0 v	V_{CC}		16.5	mA
3 T_A = -55°C	V_{OL}	3007	79 80 81 82 83 84	2.1 v	12 mA	2.1 v	12 mA	2.1 v	12 mA		12 mA	2.1 v	12 mA	2.1 v	12 mA	2.1 v	4.5 v	1Y 2Y 3Y 4Y 5Y 6Y		0.4	Vdc
	V_{OH}	3006	85 86 87 88 89 90	1.4 v	-0.12mA	1.4 v	-0.12mA	1.4 v	-0.12mA		-0.12mA	1.4 v	-0.12mA	1.4 v	-0.12mA	1.4 v		1Y 2Y 3Y 4Y 5Y 6Y	3.3		
	I_{OS}	3011	91 92 93 94 95 96	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	GND	GND	5.5 v	1Y 2Y 3Y 4Y 5Y 6Y	-.67	-1.34	mA
	I_{IH}	3010	97 98 99 100 101 102	4.0 v		4.0 v		4.0 v				4.0 v		4.0 v		4.0 v		1A 2A 3A 4A 5A 6A		2.0	µA

See footnote at end of table.

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TABLE III. Group A inspection for device type 04. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Test limits Max	Unit
				1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	VCC				
1 T_A=25°C	V_{OL}	3007	1	1.9 V	1.9 V	12 mA	1.9 V	1.9 V	12 mA	GND	12 mA	1.9 V	1.9 V	12 mA	1.9 V	1.9 V	4.5 V	1Y		0.4	Vdc
			2															2Y			
			3															3Y			
			4															4Y			
	V_{OH}	3006	5	1.1 V	1.1 V	-0.12mA	1.1 V	1.1 V	-0.12mA		-0.12mA	1.1 V	1.1 V	-0.12mA	1.1 V			1Y	3.4		
			6			-0.12mA			-0.12mA									1Y			
			7															2Y			
			8															2Y			
			9															3Y			
			10															3Y			
			11															4Y			
			12															4Y			
	I_{CS}	3011	13	GND		GND	GND		GND	GND	GND	GND		GND	GND		5.5 V	1Y	-.67	-1.34	mA
			14															2Y			
			15															3Y			
			16															4Y			
	I_{IH}	3010	17	4.0 V	GND	GND	4.0 V	GND				4.0 V	GND			GND		1A		2	µA
			18	GND	4.0 V		GND	4.0 V				GND	4.0 V			4.0 V		1B			
			19															2A			
			20															2B			
			21															3A			
			22															3B			
			23															4A			
			24															4B			
	I_{IL}	3009	25	0.4 V	4.0 V		0.4 V	4.0 V				0.4 V	4.0 V		0.4 V	4.0 V		1A	-.6	-1.50	mA
			26	4.0 V	0.4 V		4.0 V	0.4 V				4.0 V	0.4 V		4.0 V	0.4 V		1B			
			27															2A			
			28															2B			
			29															3A			
			30															3B			
			31															4A			
			32															4B			
	I_{CEX}	3005	33	1.1 V		4.5 V	1.1 V		4.5 V		4.5 V	1.1 V		4.5 V	1.1 V		4.5 V	1Y		50	µA
			34															2Y			
			35															3Y			
			36															4Y			
	I_{CCL}	3005	37														5.0 V	VCC		13	mA
	I_{CCH1}		38				GND					GND			GND		5.0 V	VCC		5.88	mA
	I_{CCH2}		39				GND					GND			GND		6.0 V	VCC		11	mA
3 T_A=125°C	V_{OL}	3007	40	1.7 V	1.7 V	12 mA	1.7 V	1.7 V	12 mA		12 mA	1.7 V	1.7 V	12 mA	1.7 V	1.7 V	4.5 V	1Y		0.45	Vdc
			41															2Y			
			42															3Y			
			43															4Y			
	V_{OH}	3008	44	0.8 V	0.8 V	-0.12mA	0.8 V	0.8 V	-0.12mA		-0.12mA	0.8 V	0.8 V	-0.12mA	0.8 V	0.8 V		1Y	2.85		
			45															1Y			
			46															2Y			
			47															2Y			
			48															3Y			
			49															3Y			
			50															4Y			
			51															4Y			

See footnote at end of table.

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Max	Unit
				1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}				
2 T_A = 125°C	I_{OS}	3011	52	GND		GND				GND							5.5 v	1Y	-.59	-1.3	mA
			53				GND		GND									2Y			
			54								GND	GND						3Y			
			55											GND	GND			4Y			
	I_{IH}	3010	56	4.0 v	GND													1A		5.0	µA
			57	GND	4.0 v													1B			
			58				4.0 v	GND										2A			
			59				GND	4.0 v										2B			
			60									4.0 v	GND					3A			
			61									GND	4.0 v					3B			
			62												4.0 v	GND		4A			
			63												GND	4.0 v		4B			
	I_{IL}	3009	64	0.4 v	4.0 v													1A	-.6	-1.50	mA
			65	4.0 v	0.4 v													1B			
			66				0.4 v	4.0 v										2A			
			67				4.0 v	0.4 v										2B			
			68									0.4 v	4.0 v					3A			
			69									4.0 v	0.4 v					3B			
			70												0.4 v	4.0 v		4A			
			71												4.0 v	0.4 v		4B			
	I_{CEX}		72	0.8 v		4.5 v											4.5 v	1Y		100	µA
			73				0.8 v		4.5 v									2Y			
			74								4.5 v	0.8 v						3Y			
			75											4.5 v	0.8 v			4Y			
	I_{CCL}	3005	76														5.0 v	V_{CC}		13	mA
	I_{CCH1}		77	GND			GND					GND			GND		5.0 v	V_{CC}		5.88	mA
	I_{CCH2}		78	GND			GND					GND			GND		5.0 v	V_{CC}		11	mA
3 T_A = -55°C	V_{OL}	3007	79	2.1 v	2.1 v	12 mA											4.5 v	1Y		0.4	Vdc
			80				2.1 v	2.1 v	12 mA									2Y			
			81								12 mA	2.1 v	2.1 v					3Y			
			82											12 mA	2.1 v	2.1 v		4Y			
	V_{OH}	3006	83	1.4 v		-0.12mA												1Y	3.3		
			84		1.4 v	-0.12mA												1Y			
			85				1.4 v		-0.12mA									2Y			
			86					1.4 v	-0.12mA									2Y			
			87								-0.12mA	1.4 v						3Y			
			88								-0.12mA		1.4 v					3Y			
			89											-0.12mA	1.4 v			4Y			
			90											-0.12mA		1.4 v		4Y			
	I_{OS}	3011	91	GND		GND											5.5 v	1Y	-.67	-1.34	mA
			92				GND		GND									2Y			
			93								GND	GND						3Y			
			94											GND	GND			4Y			
	I_{IH}	3010	95	4.0 v	GND													1A		2.0	µA
			96	GND	4.0 v													1B			
			97				4.0 v	GND										2A			
			98				GND	4.0 v										2B			
			99									4.0 v	GND					3A			
			100									GND	4.0 v					3B			
			101												4.0 v	GND		4A			
			102												GND	4.0 v		4B			

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1B	3 1Y	4 2A	5 2B	6 2Y	7 GND	8 3Y	9 3A	10 3B	11 4Y	12 4A	13 4B	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A= -55°C	I_{IL}	3009	103	0.4 v	4.0 v					GND							5.5 v	1A	-.6	-1.50	mA
			104	4.0 v	0.4 v													1B			
			105				0.4 v	4.0 v										2A			
			106				4.0 v	0.4 v										2B			
			107									0.4 v	4.0 v					3A			
			108									4.0 v	0.4 v					3B			
			109												0.4 v	4.0 v		4A			
			110												4.0 v	0.4 v		4B			
	I_{CEX}		111	1.4 v		4.5 v			4.5 v								4.5 v	1Y		50	µA
			112				1.4 v				4.5 v	1.4 v						2Y			
			113															3Y			
			114											4.5 v	1.4 v			4Y			
	I_{CCL}	3005	115														5.0 v	V_{CC}		13	mA
	I_{CCH1}		116	GND			GND					GND			GND		5.0 v	V_{CC}		5.88	mA
	I_{CCH2}		117	GND			GND					GND			GND		5.0 v	V_{CC}		11	mA
9 T_A=25°C	t_{PHL}	3003 (Fig. 5)	118	IN		OUT											5.0 v	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	10	30	ns
			119				IN		OUT		OUT	IN									
			120											OUT	IN						
			121																		
	t_{PLH}		122	IN		OUT												1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	25	80	
			123				IN		OUT		OUT	IN									
			124											OUT	IN						
			125																		
10 T_A=125°C	t_{PHL}		126	IN		OUT												1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	10	35	
			127				IN		OUT		OUT	IN									
			128											OUT	IN						
			129																		
	t_{PLH}		130	IN		OUT												1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	25	112	
			131				IN		OUT		OUT	IN									
			132											OUT	IN						
			133																		
11 T_A= -55°C	t_{PHL}		134	IN		OUT												1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	10	40	
			135				IN		OUT		OUT	IN									
			136											OUT	IN						
			137																		
	t_{PLH}		138	IN		OUT												1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	25	80	
			139				IN		OUT		OUT	IN									
			140											OUT	IN						
			141																		

1/ Pins not designated may be HIGH level logic, LOW level logic or open.

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TABLE III. Group A inspection for device type 05. 1/

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1 1A	2 1B	3 2A	4 2B	5 2C	6 2Y	7 GND	8 3Y	9 3A	10 3B	11 3C	12 1Y	13 1C	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A=25°C	V_{OL}	3007	1	1.9 v	1.9 v					GND					12 mA	1.9 v	4.5 v	1Y		0.4	Vdc
			2			1.9 v	1.9 v	1.9 v	12 mA									2Y			
			3								12 mA	1.9 v	1.9 v	1.9 v				3Y			
	V_{OH}	3006	4	1.1 v											-0.12 mA			1Y	2.4		
			5		1.1 v													1Y			
			6													1.1 v		1Y			
			7			1.1 v			-0.12 mA									2Y			
			8				1.1 v											2Y			
			9					1.1 v										2Y			
			10								-0.12mA	1.1 v						3Y			
			11										1.1 v					3Y			
			12											1.1 v				3Y			
	I_{OS}	3011	13												GND	GND	5.5 v	1Y	-.67	-1.34	mA
			14					GND	GND									2Y			
			15								GND			GND				3Y			
	I_{IH}	3010	16	4.0 v	GND											GND		1A		2.0	µA
			17	GND	4.0 v											GND		1B			
			18	GND	GND											4.0 v		1C			
			19			4.0 v	GND	GND										2A			
			20			GND	4.0 v	GND										2B			
			21			GND	GND	4.0 v										2C			
			22									4.0 v	GND	GND				3A			
			23									GND	4.0 v	GND				3B			
			24									GND	GND	4.0 v				3C			
	I_{IL}	3009	25	0.4 v	4.0 v											4.0 v		1A	-.6	-1.50	mA
			26	4.0 v	0.4 v											4.0 v		1B			
			27	4.0 v	4.0 v											0.4 v		1C			
			28			0.4 v	4.0 v	4.0 v										2A			
			29			4.0 v	0.4 v	4.0 v										2B			
			30			4.0 v	4.0 v	0.4 v										2C			
			31									0.4 v	4.0 v	4.0 v				3A			
			32									4.0 v	0.4 v	4.0 v				3B			
			33									4.0 v	4.0 v	0.4 v				3C			
	I_{CEX}		34	1.1 v											4.5 v			1Y		50	µA
			35			1.1 v			4.5 v								4.5 v	2Y			
			36								4.5 v	1.1 v						3Y			
	I_{CCL}	3005	37														5.0 v	V_{CC}		8.75	mA
	I_{CCH1}		38	GND		GND						GND					5.0 v	V_{CC}		4.41	mA
	I_{CCH2}		39	GND		GND						GND					5.0 v	V_{CC}		8.25	mA
2 T_A=125°C	V_{OL}	3007	40	1.7 v	1.7 v										12 mA	1.7 v	4.5 v	1Y		0.45	Vdc
			41			1.7 v	1.7 v	1.7 v	12 mA									2Y			
			42								12 mA	1.7 v	1.7 v	1.7 v				3Y			
	V_{OH}	3006	43	0.8 v											-0.12 mA			1Y	2.85		
			44		0.8 v													1Y			
			45													0.8 v		1Y			
			46			0.8 v			-0.12 mA									2Y			
			47				0.8 v											2Y			
			48					0.8 v										2Y			
			49								-0.12mA	0.8 v						3Y			
			50										0.8 v					3Y			
			51											0.8 v				3Y			

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TABLE II. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Max	Unit
				1A	1B	2A	2B	2C	2Y	GND	3Y	3A	3B	3C	1Y	1C	VCC				
2 T_A=125°C	I_{OS}	3011	52					GND	GND	GND					GND	GND	5.5 v	1Y	-.59	-1.3	mA
			53								GND			GND				2Y			
			54															3Y			
	I_{IH}	3010	55	4.0 v	GND											GND		1A		5	µA
			56	GND	4.0 v											GND		1B			
			57	GND	GND											4.0 v		1C			
			58			4.0 v	GND	GND										2A			
			59			GND	4.0 v	GND										2B			
			60			GND	GND	4.0 v										2C			
			61									4.0 v	GND	GND				3A			
			62									GND	4.0 v	GND				3B			
			63									GND	GND	4.0 v				3C			
	I_{IL}	3009	64	0.4 v	4.0 v											4.0 v		1A	-.8	-1.50	mA
			65	4.0 v	0.4 v											4.0 v		1B			
			66	4.0 v	4.0 v											0.4 v		1C			
			67			0.4 v	4.0 v	4.0 v										2A			
			68			4.0 v	0.4 v	4.0 v										2B			
			69			4.0 v	4.0 v	0.4 v										2C			
			70									0.4 v	4.0 v	4.0 v				3A			
			71									4.0 v	0.4 v	4.0 v				3B			
			72									4.0 v	4.0 v	0.4 v				3C			
	I_{CEX}		73	0.8 v											4.5 v		4.5 v	1Y		100	µA
			74			0.8 v			4.5 v									2Y			
			75								4.5 v	0.8 v						3Y			
	I_{CCL}	3005	76														5.0 v	V_{CC}		9.75	mA
	I_{CCH1}		77	GND		GND						GND					5.0 v	V_{CC}		4.41	mA
	I_{CCH2}		78	GND		GND						GND					5.0 v	V_{CC}		8.25	mA
3 T_A= -55°C	V_{OL}	3007	79	2.1 v	2.1 v										12 mA	2.1 v	4.5 v	1Y		0.4	Vdc
			80			2.1 v	2.1 v	2.1 v	12 mA									2Y			
			81								12 mA	2.1 v	2.1 v	2.1 v				3Y			
	V_{OH}	3006	82	1.4 v											-0.12mA			1Y	3.5		
			83		1.4 v											1.4 v		1Y			
			84			1.4 v			-0.12mA									1Y			
			85				1.4 v											2Y			
			86					1.4 v										2Y			
			87								-0.12mA	1.4 v						2Y			
			88										1.4 v					3Y			
			89											1.4 v				3Y			
			90															3Y			
	I_{OS}	3011	91					GND	GND						GND	GND	5.5 v	1Y	-.67	-1.34	mA
			92								GND			GND				2Y			
			93															3Y			
	I_{IH}	3010	94	4.0 v	GND											GND		1A		2.0	µA
			95	GND	4.0 v											GND		1B			
			96	GND	GND											4.0 v		1C			
			97			4.0 v	GND	GND										2A			
			98			GND	4.0 v	GND										2B			
			99			GND	GND	4.0 v										2C			
			100									4.0 v	GND	GND				3A			
			101									GND	4.0 v	GND				3B			
			102									GND	GND	4.0 v				3C			

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C and D Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Max	Unit
				1A	1B	2A	2B	2C	2Y	GND	3Y	3A	3B	3C	1Y	1C	V_{CC}				
3 T_A= -55°C	I_{IL}	3009	103	0.4 v	4.0 v					GND						4.0 v	5.5 v	1A	-.6	-1.50	mA
			104	4.0 v	0.4 v											4.0 v		1B			
			105	4.0 v	4.0 v											0.4 v		1C			
			106			0.4 v	4.0 v	4.0 v										2A			
			107			4.0 v	0.4 v	4.0 v										2B			
			108			4.0 v	4.0 v	0.4 v										2C			
			109									0.4 v	4.0 v	4.0 v				3A			
			110									4.0 v	0.4 v	4.0 v				3B			
			111									4.0 v	4.0 v	0.4 v				3C			
	I_{CEX}		112	1.4 v											4.5 v		4.5 v	1Y		50	µA
			113			1.4 v			4.5 v									2Y			
			114								4.5 v	1.4 v						3Y			
	I_{CCL}	3005	115														5.0 v	V_{CC}		9.75	mA
	I_{CCH1}		116	GND		GND						GND					5.0 v	V_{CC}		4.41	mA
	I_{CCH2}		117	GND		GND						GND					5.0 v	V_{CC}		8.25	mA
9 T_A=25°C	t_{PHL}	3003 (Fig. 8)	118	IN		IN			OUT						OUT		5.0 v	1A to 1Y	10	30	ns
			119															2A to 2Y			
			120								OUT	IN						3A to 3Y			
	t_{PLH}		121	IN		IN			OUT						OUT			1A to 1Y	25	80	
			122															2A to 2Y			
			123								OUT	IN						3A to 3Y			
10 T_A=125°C	t_{PHL}		124	IN		IN			OUT						OUT			1A to 1Y	10	35	
			125															2A to 2Y			
			126								OUT	IN						3A to 3Y			
	t_{PLH}		127	IN		IN			OUT						OUT			1A to 1Y	25	115	
			128															2A to 2Y			
			129								OUT	IN						3A to 3Y			
11 T_A= -55°C	t_{PHL}		130	IN		IN			OUT						OUT			1A to 1Y	10	40	
			131															2A to 2Y			
			132								OUT	IN						3A to 3Y			
	t_{PLH}		133	IN		IN			OUT						OUT			1A to 1Y	25	80	
			134															2A to 2Y			
			135								OUT	IN						3A to 3Y			

1/ Pins not designated may be HIGH level logic, LOW level logic or open.

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4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced to room temperature within 30 minutes after removal of the test condition.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with MIL-M-38510, except that the rough handling test shall not apply.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging and packing. Microcircuits shall be prepared for delivery in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.3 Ordering data. The contract or order should specify the following:

(a) Complete part number (see 1.2).
(b) Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
(c) Requirement for certificate of compliance, if applicable.
(d) Requirements for notification of change of product or process to procuring activity in addition to notification to qualifying activity, if applicable.
(e) Requirements for packaging and packing.
(f) Requirements for failure analysis (including required test condition of Method 5003), corrective action and reporting of results, if applicable.
(g) Requirements for product assurance options.
(h) Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchases by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
I_{CEX}	Collector cutoff current
V_{IN}	Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured to Government logistic support will be procured to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.6 Substitutability. Microcircuits covered by this specification are substitutable for the following commercial device types:

Device type	Commercial type
01	930
02	MIC935 and MC940
03	936
04	946
05	962

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MI
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - WC, MU
Navy - OS, AS, SH, CG
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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